Multiple linear regression - Cholesterol

MHEDAS

Biomedical Statistics

Multiple linear regression

Outcome: cholesterol (C) 📈

Explanatory variables: height in cm (H); Weight in Kg (W); Age in years (A)

Main concepts: confounders and Collinearity

Load libraries

```
library(car) # linear modeling

Loading required package: carData

library(dm) # data display

Loading required package: lattice

Loading required package: grid

Loading required package: latticeExtra

Loading required package: multcomp

Loading required package: mvtnorm

Loading required package: sandwich

Loading required package: survival

Loading required package: TH.data

Loading required package: MASS

Attaching package: 'TH.data'

The following object is masked from 'package:MASS':

    geyser

Loading required package: grDevices

Attaching package: 'dm'

The following objects are masked from 'package:car':

    logit, vif

The following object is masked from 'package:base':

    cut.P
```

Read data

```
dd <- read.csv2("cholesterol.csv") # dataset
head(dd,3)                         # table view
summ(dd)

  id  C  H  W  A
1  1 270 64 71 23
2  2 215 63 61 20
3  3 215 63.5 61 20.00
4  4 215 64.5 61 20.00
5  5 215 64.5 61 20.00
6  6 215 64.5 61 20.00
```

Descriptive plot

```
## package sjplot
sjp(C ~ W, bootstrap=F, smooth=F, dd)
```

relationship looks wrong (weight, cholesterol) -> check for confounders

Null model

```
m0M <- lm(C~1,dd) #intercept model
summary(m0M)

Call:
lm(formula = C ~ 1, data = dd)

Residuals:
    Min      1Q  Median      3Q     Max
-55.00 -20.00  -1.00  20.00 200.00

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  215.00   1.000  215.0   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 31.61 on 99 degrees of freedom

confint(m0M,level=0.95)
                2.5 %   97.5 %
(Intercept) 213.0202 216.9798
```

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```

Predictions

```
dd <- predict(m0M, interval="confidence", stderr=T) # Expected value, average of all
dd <- predict(m0M, interval="prediction", stderr=T) # Predicted value, individual, one individuals

Warning in predict.lm(m0M, interval = "prediction", stderr = T): prediction is current
```

~ Plot

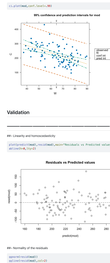
```
# ~ ggplot
plot(mpg.ggpl,aes(x=hwy,y=cty),pch=1,cex=2)
# ggplot(mpg,aes(x=hwy,y=cty))+geom_point()  # Points

lines(ANCOVPC_CFB,"fit",lty=1,col="black")  # Point estimate

lines(ANCOVPC_CFB,"fit",lty=2,col="black")  # Lower limit confidence interval
lines(ANCOVPC_CFB,"fit",lty=2,col="black")  # Upper limit confidence interval

lines(ANCOVPC_CFB,"fit",lty=2,col="red")    # Lower limit prediction interval
lines(ANCOVPC_CFB,"fit",lty=2,col="red")    # Upper limit prediction interval
```

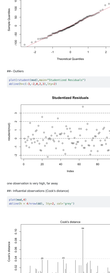
95% prediction bands



~ Alternative plot

```
ci.plot(mod,conf.level=0.95)
```

95% confidence and prediction intervals for mod



Validation

```
## Linearity and homoscedasticity
plot(mod,which=1,main="Residuals vs Predicted values")
abline(h=0,lty=2)
```

Residuals vs Predicted values

```
## Normality of the residuals
qqnorm(rstandard(mod))
qqline(rstandard(mod),col=2)
```

Normal Q-Q Plot

```
## Outliers
plot(rstudent(mod),main="Studentized Residuals")
abline(h=3,lty=2),abline(h=-3,lty=2)
```

Studentized Residuals

one observation is very high, far away

```
## Influential observations (Cook's distance)
plot(mod,4)
abline(h = 8/(n-2*k), lty=2, col="grey")
```

Cook's distance

one observation has a very high influence

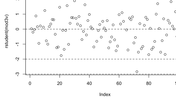
```
## Influential observations (dffits is an alternative measure of the influence)
# or
plot(dffits(mod),main="dffits")
abline(h=3/sqrt(n),lty=2),abline(h=-3/sqrt(n),lty=2)
```

dffits


```
## OR in one
cbind(OR=exp(C),CI)
print(mod)
```

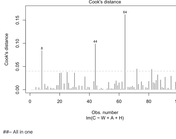

Problem with the interpretation

```
checking collinearity BM ~~~
```



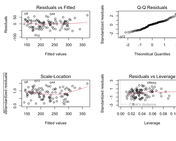
##- Influential observations (Cook's distance)

```
plot(mod2v,4)
abline(h = 4/nrow(df), lty=2, col='grey')
```



##- All in one

```
par(mfrow=c(2,2))
plot(mod2v)
```



Dealing with collinearity –> Create variable EW: excess weight

defined by expert: ## Ideal weight = 0.5 * height - 10 ##
New variable include info of weight and height

```
real_weight <- df$W
ideal_weight <- df$H/2 - 10
df$EW <- pmax(0, real_weight - ideal_weight)
```

Multiple linear regression with excess weight

```
mod2v <- lm(C ~ EW + A, df)
summary(mod2v)
```

```
Call:
lm(formula = C ~ EW + A, data = df)

Residuals:
    Min      1Q  Median      3Q     Max 
-74.838 -21.208   1.658  20.369  65.293 

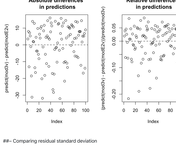
Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 386.8560   12.9274  29.925  < 2e-16 ***
EW            9.9347    1.0573   9.397 2.09e-15 ***
A           -12.4542    0.8379 -14.863  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 29.93 on 97 degrees of freedom
Multiple R-squared:  0.7654,	Adjusted R-squared:  0.7606 
F-statistic: 158.2 on 2 and 97 DF,  p-value: < 2.2e-16
```

Comparing predictions

##- Comparing (absolute/relative) predictions

```
par(mfrow=c(1,2))
plot(prediction(mod2v) - predict(mod2v),
     main="Absolute differences \n in predictions")
abline(h=0,lty=2)
plot((prediction(mod2v) - predict(mod2v))/predict(mod2v),
     main="Relative differences \n in predictions")
abline(h=0,lty=2)
```



##- Comparing residual standard deviation

```
(res_sd <- c(summary(mod2v)$sigma,
             summary(mod2v)$sigma))
```

```
[1] 31.48587 29.92958
```

##- Comparing coefficients

```
rbind(coef(mod2v),c(coef(mod2v),NA))
```

```
     (Intercept)        W        A       H
[1,]    386.8560 9.93468 -12.4542      NA
[2,]    386.8560 9.934729 -12.4542      NA
```

NA = independence; solved problem of collinearity

##- Tests significance

```
anova(mod2v)   # Sequential contribution
```

```
Analysis of Variance Table

Response: C
          Df Sum Sq Mean Sq F value    Pr(>F)    
EW         1  85598   85598  95.558 4.097e-16 ***
A          1 197082  197082 220.006 < 2.2e-16 ***
Residuals 97  86898     896                      
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mod2v)   # Marginal contribution
```

```
Anova Table (Type II tests)

Response: C
           Sum Sq Df F value    Pr(>F)    
EW         79096  1  88.298 2.09e-15 ***
A          98982  1 220.006 < 2.2e-16 ***
Residuals  86898 97                      
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

##- Collinearity

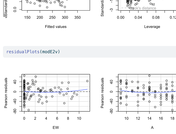
```
vif(mod2v)
```

```
       EW         A 
1.000639 1.000639 
```

Validation

##- All in one

```
par(mfrow=c(2,2))
plot(mod2v)
```




```
residualPlots(mod2v)
```

```
           Test stat Pr(>|Test stat|)
EW            0.5052        0.5531
A             1.3236        0.1888
Tukey test    1.5569        0.1195
```

Predictions

##- Prediction according to age for patients without excess of weight

```
age <- 9:20
C1 <- data.frame(cbind(A=age,EW=0))          # Individual characteristics
CC1 <- predict(mod2v, C1, interval="confidence") # 95% Confidence intervals
CP1 <- predict(mod2v, C1, interval="prediction") # 95% Prediction intervals

##-- Plot
par(mfrow=c(1,1))
plot(age,CC1[,"fit"],ty="l",ylim=c(min(CP1),max(CP1)),
     main="95% Prediction band for cholesterol 95% w/o excess of weight")

lines(age,CC1[,"lwr"],lty=2,col="blue")
lines(age,CC1[,"upr"],lty=2,col="blue")

lines(age,CP1[,"lwr"],lty=2,col="red")
lines(age,CP1[,"upr"],lty=2,col="red")
```

blue = average with 95% confidence for all people in the world

##- Prediction for a patient with W=65, A=15 (H=150)

```
C0 <- data.frame(cbind(EW=c(max(0,65-15*0.5-10)), A=c(15)))
predict(mod2v, C0, interval="confidence", level=.95) # confidence interval
```

```
       fit     lwr     upr
1 200.043 192.6502 207.4278
```

```
predict(mod2v, C0, interval="prediction", level=.95) # prediction interval
```

```
       fit     lwr     upr
1 200.043 140.189 259.902
```

Contrasts

##- Contrast if E[C]>150 for the previous individual with W=65, A=15 (H=150 option. 0 = see if value could be 150 -> no cause it is a outside CI # 1st option: manual

```
pC <- predict(mod2v,C0,se.fit=T)
2*(1 - pt(qC1f1-150/pCse.fit, pC$df)) # p value
```

```
        1
0.000201667
```

<0.05 = significant difference to predicted average (200.043)

2nd option: "linearHypothesis" function

```
linearHypothesis(mod2v,c(1,0,15),150)
```

```
Linear hypothesis test:
(Intercept)  + 15 A = 150

Model 1: restricted model
Model 2: C ~ EW + A

  Res.Df  RSS DF Sum of Sq      F  Pr(>F)
1     98 93416
2     97 86898  1    6526.1 7.2854 0.008202 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

3rd option: "lht" function

```
lht(mod2v,c(1,0,15),150)
```

```
Linear hypothesis test:
(Intercept)  + 15 A = 150

Model 1: restricted model
Model 2: C ~ EW + A

  Res.Df  RSS DF Sum of Sq      F  Pr(>F)
1     98 93416
2     97 86898  1    6526.1 7.2854 0.008202 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(qC1f1-150)/pCse.fit)^2
```

```
        1
7.285423
```